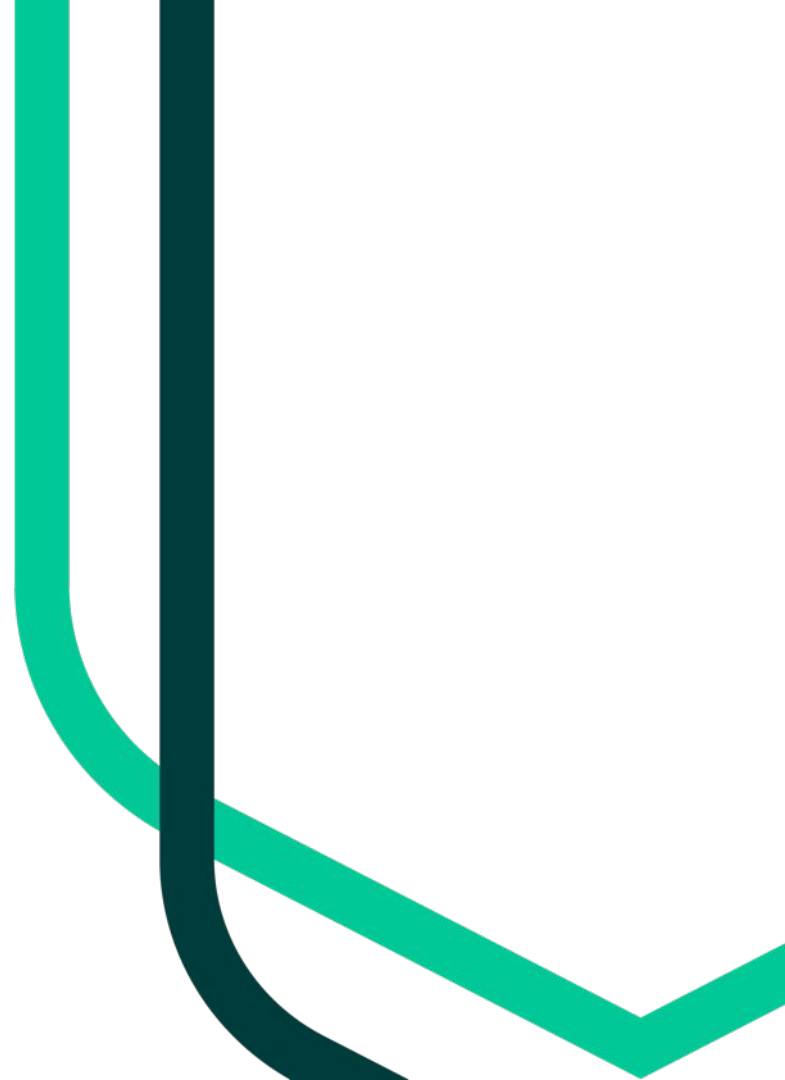




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Agentic Climate Data Communication

Arjun Biswas, Hywel .T.P. Williams, Hang Dong, Hailun Xie,
Anrijs Abele, Fai Fung



Agentic AI

Agent : A system that completes a very specific task over a period of time

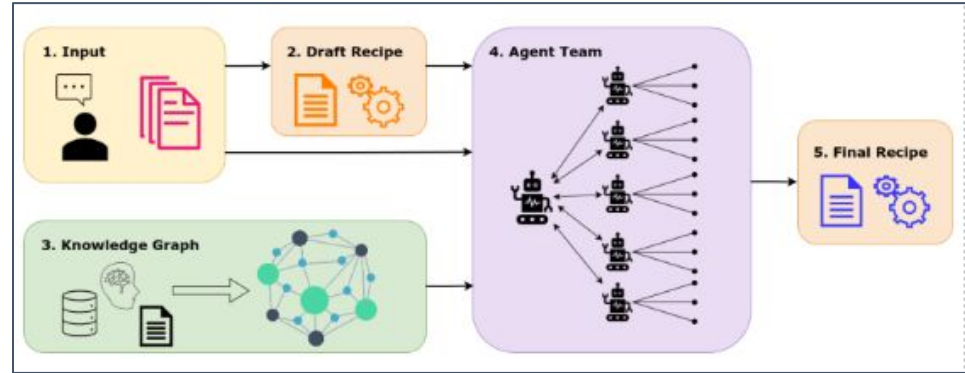
Agentic AI includes the class of **autonomous AI** systems that undertake to finish a set of complex tasks that span over long periods of time without human supervision

Agentic AI uses **tools** to communicate with external information that can be helpful to complete a particular task

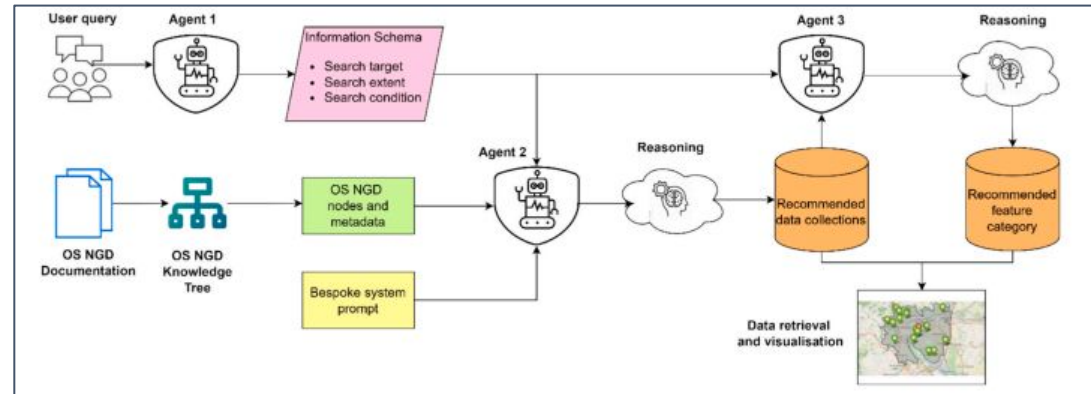


Agentic AI for different domains

- Agentic AI can be used to interact with data of different domains or organizations.
- These systems can be single agent architectures or multi-agent architectures
- Each of these agents have been tuned with domain specific information



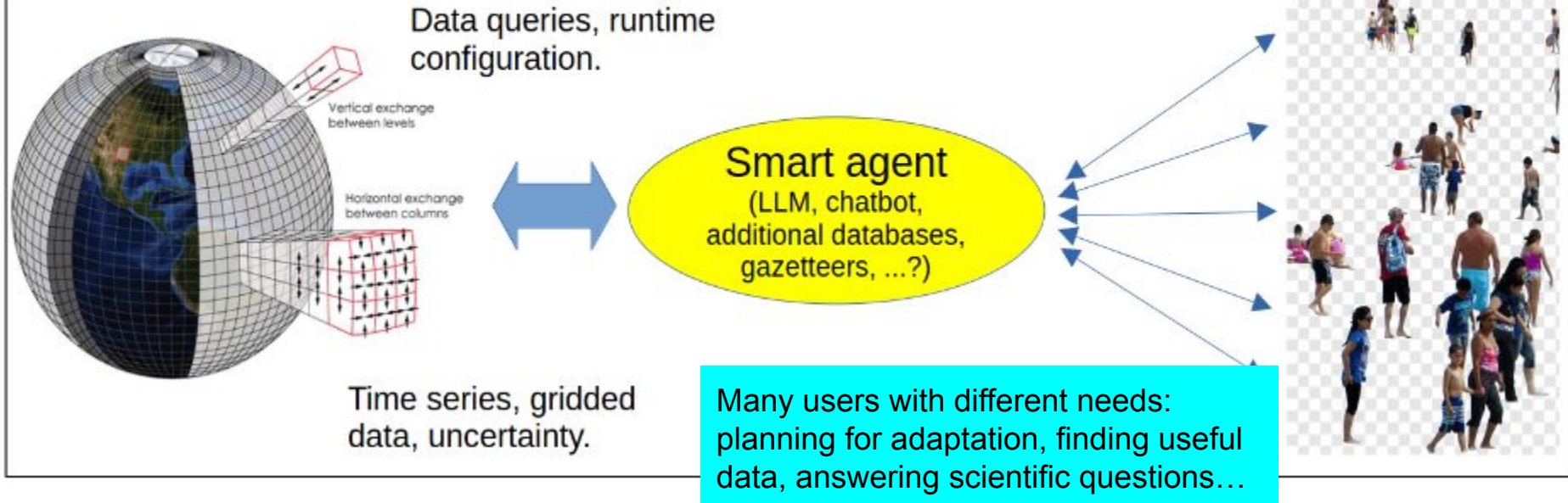
Agentic system for climate service recipe



Agentic system for geospatial information

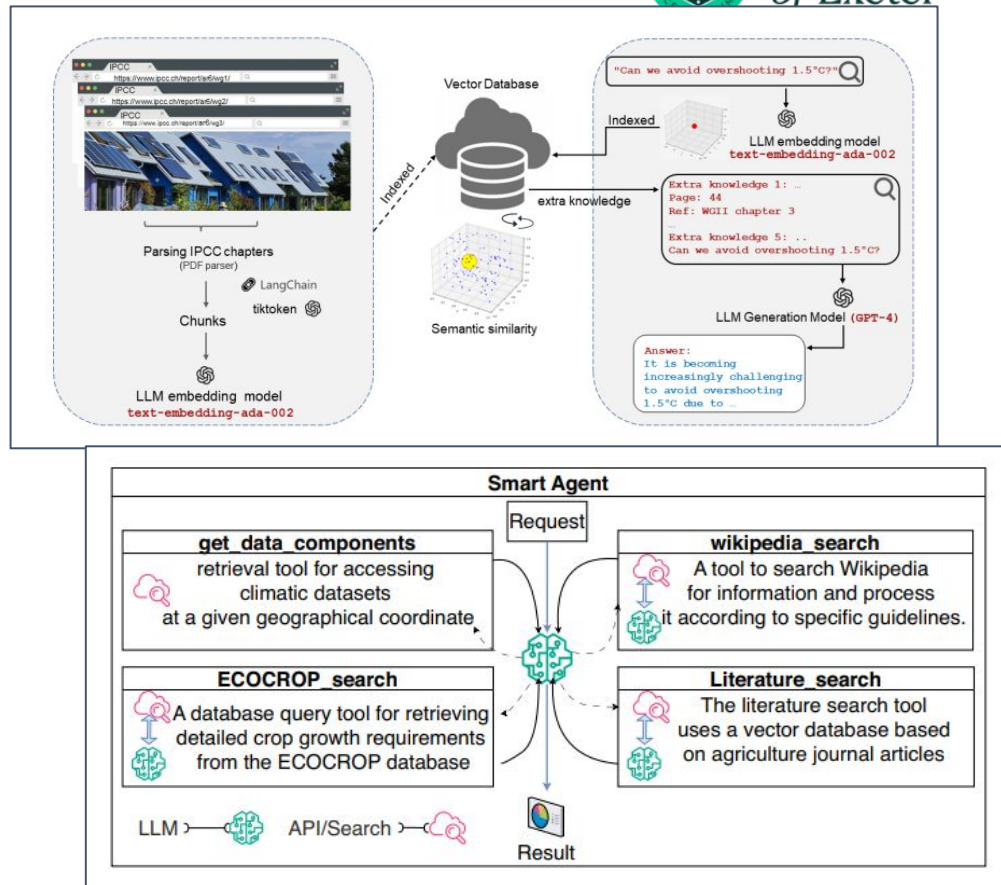
Agentic AI for climate change

How can we use Large Language Models and generative AI to connect data/models to users? At scale?



Background Literature

- Climate chatbots like ChatClimate, ClimateGPT, CPIQA use only the climate documents to answer a question, but do not use raw climate data.
- Other tools like ClimateSight use climate data, soil information but use LLM to create reasoning and inferences. These may not be suitable for different use cases which require specific data.



Article Type: Research Article

U.K. Climate Projections: Summer Daytime and Nighttime Urban Heat Island Changes in England's Major Cities

Y. T. Eunice Lo, Daniel M. Mitchell, Sylvia I. Bohnenstengel, Mat Collins, Ed Hawkins, Gabriele C. Hegerl, Manoj Joshi, and Peter A. Stott

Online Publication: 23 Sep 2020

Print Publication: 15 Oct 2020

DOI: <https://doi.org/10.1175/JCLI-D-19-0961.1>

Page(s): 9015–9030

Cited by: 33

Article History

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Urban heat Islands :

1. An urban heat island is characterized by higher near-surface air or surface skin temperatures in a given urban area compared to its rural surroundings
2. Uses max, min, mean temperature variables for the analysis at 12 km spatial resolution and daily data

Wildfire Article main features :

1. Fire Weather Index from multiple variables: maximum air temperature, mean relative humidity, mean wind speed, and precipitation.
2. The calculation of the FWI is quite complex, but it uses raw input data rather than modelling.



Natural Hazards and Earth System Sciences

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Article

Articles / Volume 22, Issue 2 / NHESS, 22, 559–575, 2022

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Research article | ©

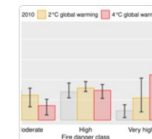
Past and future trends in fire weather for the UK

Matthew C. Perry, Emilie Vanvyve, Richard A. Betts, and Erika J. Palin

Abstract

Article Assets Peer review Metrics Related articles

21 Feb 2022



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Short summary

In the past, wildfires in the UK have occurred mainly in spring, with occasional events during...

UKCP - United Kingdom Climate Projections



APPROACH

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product

Product Selection

32 products selected

Clear all

Type

- Observations (4)
- Projections (28)

Collection

- Land observations (4)
- Land projections: global (60km) (4)
- Land projections: probabilistic (25km) (8)
- Land projections: regional (12km) (4)
- Land projections: local (2.5km) (5)
- Marine projections (7)

Scenario

- RCP 2.6 (15)
- RCP 4.5 (15)
- RCP 6.0 (5)
- RCP 8.5 (5)
- SSP5-A1B (8)

Output

- Data only (15)
- Graphs (11)
- Maps (6)

Climate Change Type

- Absolute values (14)
- Anomaly values (22)

Products

The list of products displayed below can be filtered by selecting values for the various categories shown in the column to the left.

Click on the links below to view further information or submit a request for a given product.

Plot: PDF/CDF for probabilistic projections (25km) over UK, 1961-2100

View details Process XML Submit a request

Keywords: Projections, Land projections: probabilistic (25km), Anomaly values, RCP 2.6, RCP 4.5, RCP 6.0, RCP 8.5, SRES A1B, Graphs

Generates a plot of the Probability Density Function (PDF) or Cumulative Distribution Function (CDF) for a future change in one variable for one or more emissions scenarios. Either single year averages (monthly/seasonal/annual) for a specific year from 1961 to 2100 are available or 2030 year decadal averages for the future period only. Results are available for anomalies for a given temporal average, time and location (on a 25km grid or a regional average).

Plot: Joint probabilities of two metrics for probabilistic projections (25 km) over UK, 1961-2100

View details Process XML Submit a request

Keywords: Projections, Land projections: probabilistic (25km), Anomaly values, RCP 2.6, RCP 4.5, RCP 6.0, RCP 8.5, SRES A1B, Graphs

Generates a plot of Joint Probability of future change in two selected variables for one emissions scenario. Either single year averages (monthly/seasonal/annual) for a specific year from 1961 to 2100 are available or 2030 year decadal averages for the future period only. Results are available for anomalies for a given emissions scenario, temporal average, time, location (on a 25km or regional average).

Maps: Anomalies for probabilistic projections (25km) over UK, 1961-2100

View details Process XML Submit a request

Keywords: Projections, Land projections: probabilistic (25km), Anomaly values, RCP 2.6, RCP 4.5, RCP 6.0, RCP 8.5, SRES A1B, Maps

Generates maps of 10th, 50th and 90th percentiles of future changes over the UK for one variable from the probabilistic projections. Either single year averages (monthly/seasonal/annual) for a specific year from 1961 to 2100 are available or 2030 year decadal averages for the future period only. Results are available for anomalies for a given scenario, temporal average, time and the 25km grid or regional averages.

Maps: Probabilistic projections of climate extremes (25km) over UK, 1961-2100

View details Process XML Submit a request

Keywords: Projections, Land projections: probabilistic (25km), Absolute values, RCP 2.6, RCP 4.5, RCP 6.0, RCP 8.5, SRES A1B, Maps

Generates maps of 10th, 50th and 90th percentiles of future changes over the UK for one variable and return period from the probabilistic projections of climate extremes. Single seasons for a specific year from 1961 to 2100 are available for the future period only. Results are available as absolute future values for a given emissions scenario, specific season, time range and 25km grid cell. The data incorporates a bias correction that ensures consistency with observed return levels for the baseline period of 1961-2000.

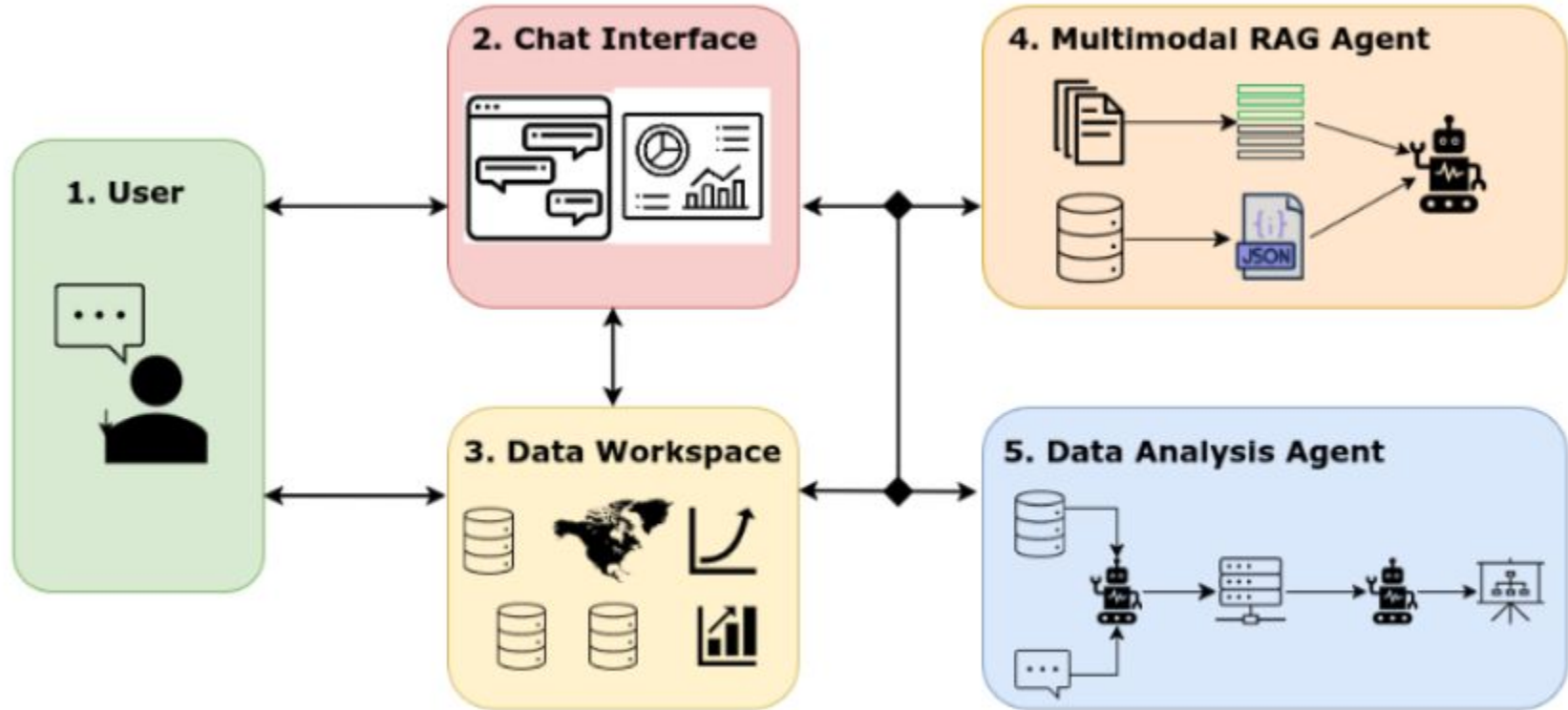
Source Data for the UKCP Chatbot



Number of Documents	Number of Images	UKCP Climate database
90	800	Database with multiple data sources

- It is a multimodal data source consisting of several documents and images.
- Apart from the documents, raw data is present in the UKCP database.
- The aim of the project is to create a chat-based interface that can answer user questions using information from various data sources.
- The most important requirement of the project was faithfulness of the answer to the source documents and preventing the spread of misinformation

Framework



Method

The entire system consists of 7 agents, each with tools that work together to complete a task

- Workflow agent : Given a question, it assigns a task to either RAG or data systems and also decides on questions that it cannot answer (zero shot prompt with dew shot examples). It is provided a record of workflow, which agent answered which question and what was the answer, based on this it can track follow up questions.
- Document Agent : It answers a question using the climate reports and climate data. It has 2 subagents. The LLM makes the choice of the tools and parameters and finally provides the answers.
 - RAG Agent : It is a single agent system that uses RAG to search the documents.
 - Climate dataAgent : Given some climate variable names it can get high resolution 60km climate data in json format.
 - Supervisor Agent : It orchestrates the overall functionality of the RAG and climate data agent

Method (contd)

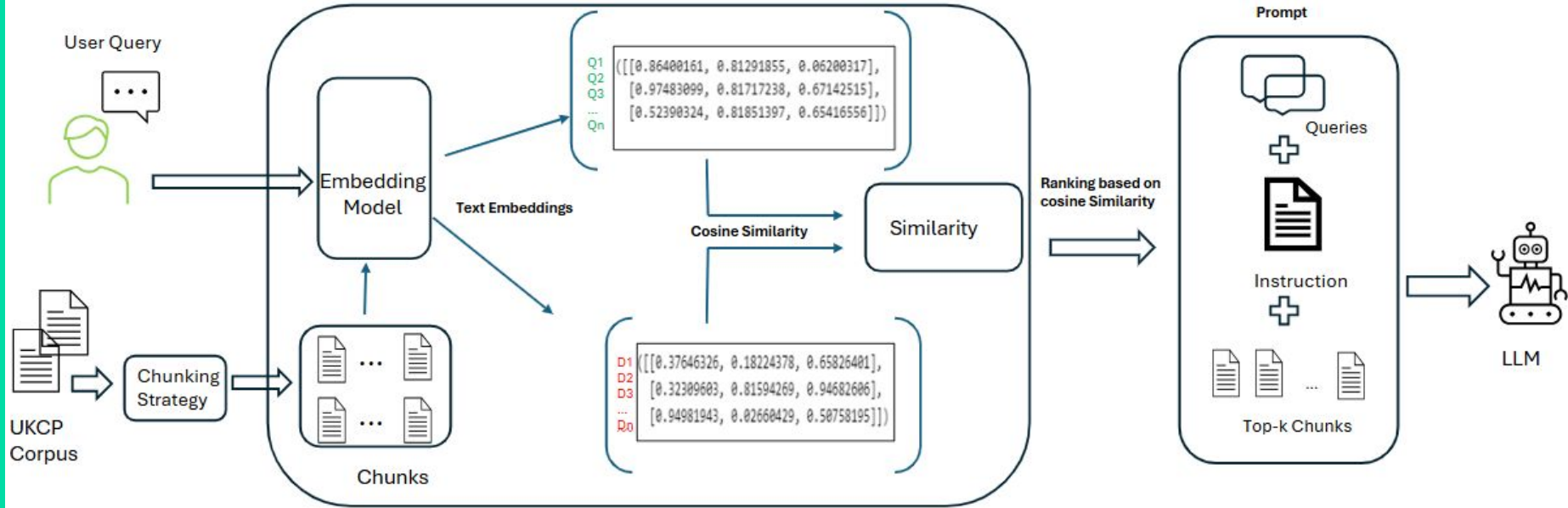
- Data Agent : Given a question, it will ask the user to provide specific parameters on the data and extract the data and perform analysis. It has 2 tools.
 - Data extraction tool : This is a specialised dynamic form that asks for parameters based on user experience then the data is extracted, the LLM validates whether the correct data is extracted continues with the analysis
 - Data Analysis Agent : This is an agent which given a question and data files, writes code on the data to run code. It has 2 agents
 - Code generator : It writes code based on the question and data
 - Code reviewer : If an error is encountered , it examines the error and provides feedback
 - Supervisor : It orchestrates the overall functionality of the data analysis agent
- Finally the LLM summarises the extracted data and the analysis

Document Module : RAG Agent

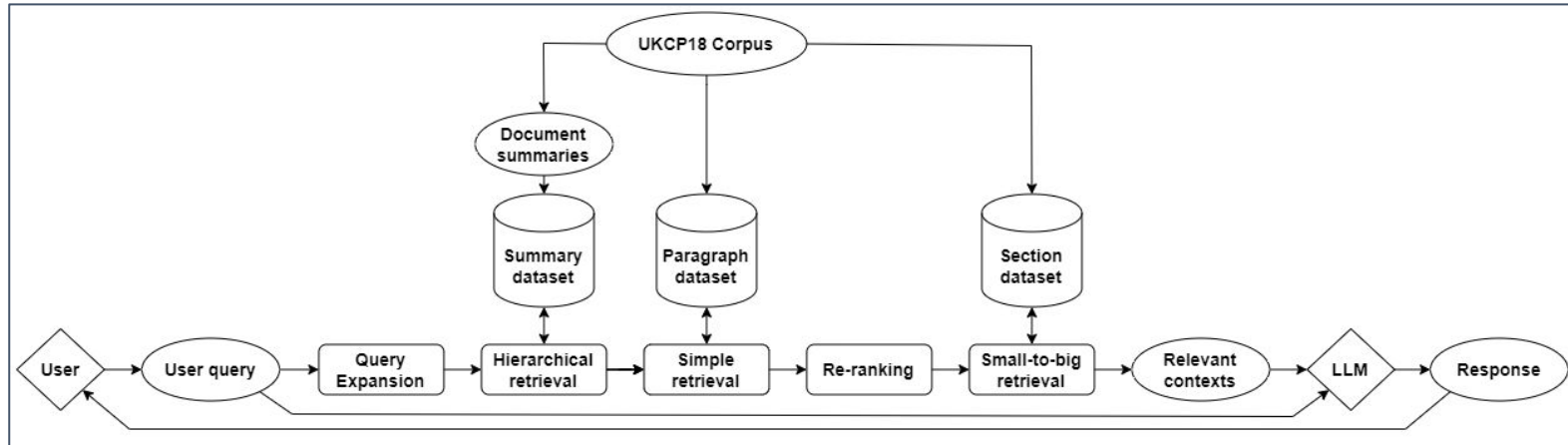


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RAG

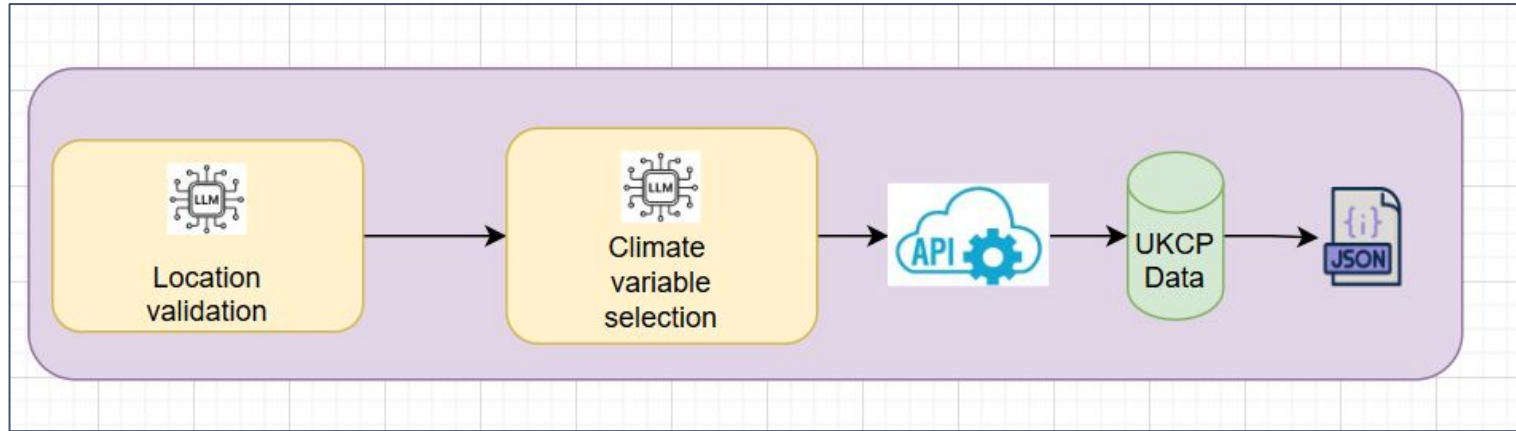


Document Module : RAG Agent (contd)



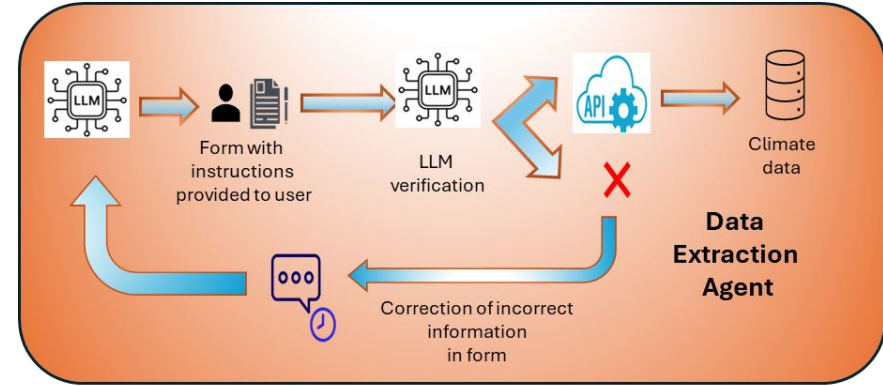
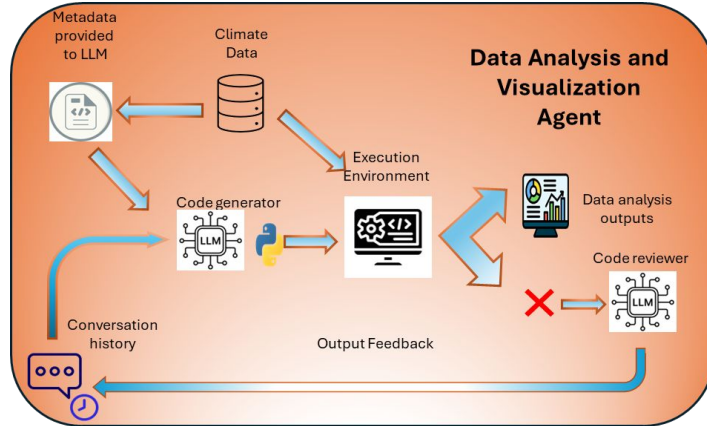
- Given a query, multiple versions are created
- Embedding model used text-embedding-ada-2
- Relevant documents are extracted using document summaries
- From each documents relevant paragraphs are identified and ranked
- Paragraphs are mapped to sections and top 10 are identified
- They are then re-ranked using a re-ranking model
- Finally the answer is summarized

Document Module : Data module (contd)



- Given a query uses LLMs to validate and map locations and climate variables to an acceptable format.
- Location is extracted from a query and converted into a bounding box of 60 km resolution
- Variable is converted into an API useable format like “tas” for mean air temperature
- Finally data is converted to JSON format

Data Module



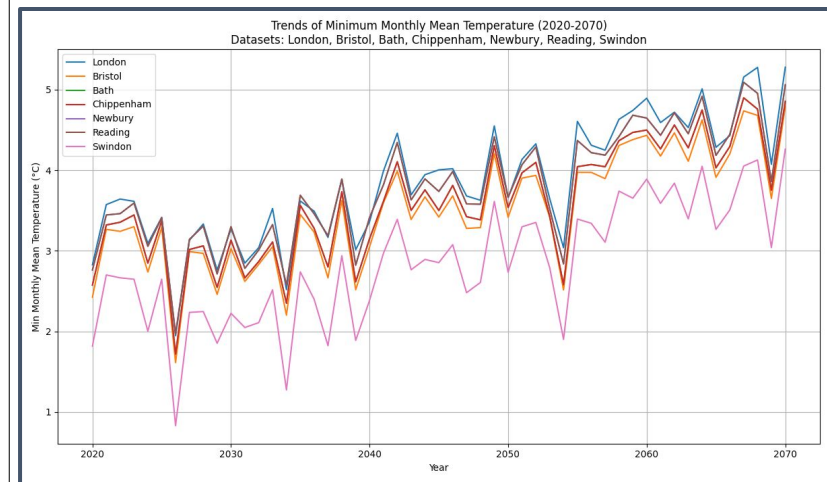
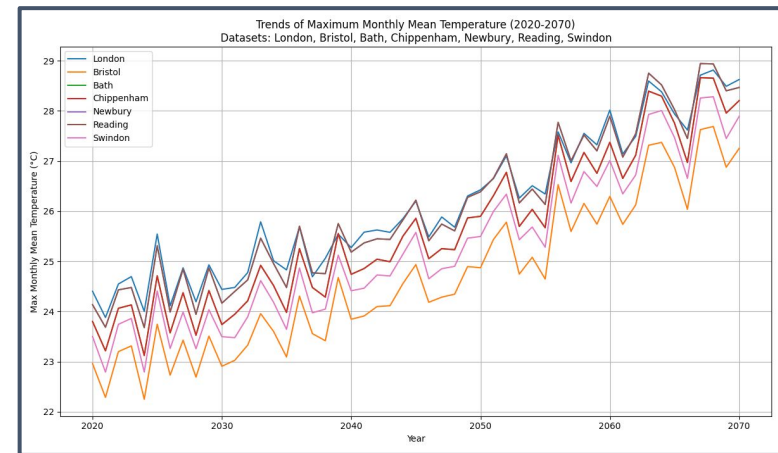
- A form is provided to the user to extract the relevant climate parameters which is then validated by the data extraction agent
- In case of incorrect information guidance is provided by the agent to the user
- Data analysis agent is designed to write climate data specific code
- The code executor tool executes the code and the results are provided to the user
- The code reviewer reviews the errors and guides the code generator agent

Results



Hybrid RAG	GPT-4o	Comment
Q2) Which part of the UK is due to experience the greatest warming under the most conservative climate projection, and what are these estimated projections?		
Response from RAG System: The UKCP18 Derived Projections for the more conservative (i.e., lower emissions/2°C global warming) scenario indicate that the South East is expected to warm the most. In this projection, the South East may see summer temperatures increase by an additional 3 to 4°C relative to the current 1981–2000 climate. Meanwhile, median warming across the UK is projected to be at least 1 to 2°C throughout the year. This means that while all parts of the UK warm, the South East stands out especially for its summer increases under this conservative projection. Response from Annual Climate Data: The annual climate data for the South East from 2020 to 2079 shows a gradual increase in temperature. For example, the temperature in 2020 was approximately 12°C, and by 2079, it is projected to rise to about 15.4°C. This data supports the RAG system's projection of significant warming in the South East, particularly during the summer months.	Under the most conservative climate projections, the UK is expected to experience the greatest warming in the northern regions, particularly in Scotland. According to the UK Climate Projections (UKCP), which provide detailed climate scenarios, the average temperature in these areas could rise by approximately 1.5 to 2 degrees Celsius by the end of the 21st century, depending on the specific emissions scenario. The projections indicate that while all parts of the UK will experience warming, the northern regions may see more significant increases due to various factors, including changes in ocean currents and atmospheric patterns. Additionally, the impacts of climate change are expected to be more pronounced in winter months, with milder winters being a notable trend. It's important to note that these projections are subject to change as new data and models are developed, but they provide a useful framework for understanding potential future climate scenarios in the UK.	Incorrect Fact by GPT 4o: UKCP18 reports consistently indicate higher summer and annual warming in the South East of England compared to Scotland under conservative emissions scenarios.

Hybrid RAG	GPT-4o	Comment
Q3) Hi, I want to use UKCP18 data to help me inform decision making about the expansion thresholds on a new powerline extending from london to bristol, to support new offshore wind power. I want to know the risk of saggingi in summer and tightening in winter given it is possible for my powerlines to be safe in a 40 degree range e.g. from -10 Celcius all the way up to 30 Celcius. Do the analysis between 2020 to 2070 and show the results by mentioning the risk years and a plot too. For the region between London and Bristol choose 5 representative regions ?		
Powerline Risk Analysis for London and Bristol (2020-2070)	Step 1: Data Acquisition - Obtain UKCP18 Data: Download the relevant temperature projections for London and Bristol and 5 representative regions from the UK Climate Projections (UKCP18) dataset. Focus on the regional climate projections for the years 2020 to 2070. Step 2: Define Temperature Thresholds - Define Safe Operating Ranges: - Option 1: Safe from -10°C to 30°C - Option 2: Safe from -5°C to 35°C Step 3: Analyze Temperature Projections - Extract Monthly Average Temperatures: For both London and Bristol, extract the monthly average temperatures for each year from 2020 to 2070. - Identify Risk Years: Step 4: Data Analysis - Count Risk Years: Count the number of years that exceed the defined thresholds for both sagging and tightening for each city. - calculate Probability: Calculate the probability of exceeding the thresholds for each option over the 50-year period.	GPT 4o does not possess capabilities to analyse data
Max Temperature Trends Overall Trend: Most locations show an increasing trend in maximum monthly mean temperatures, with some fluctuations. Risk of Sagging: No years were identified where the monthly maximum temperature exceeded 30°C, indicating a low risk of sagging due to high temperatures. Minimum Temperature Trends Overall Trend: Minimum monthly mean temperatures generally fluctuate, with some locations showing a slight warming trend, indicating less extreme cold. Risk of Tightening: No years were identified where the monthly minimum temperature dropped below -10°C, indicating a low risk of tightening due to low temperatures.		



Discussion – Evaluation

- Document Agent Evaluation:
 - RAG Evaluation
 - How relevant is the answer to the question provided by the LLM
 - Is the answer provided by the LLM based system in the source data
 - Is the answer provided by the LLM based system factually correct and complete
 - Subject Matter Experts at the Met Office to rate answers from the RAG agent
 - Evaluation of Data extraction : Is the chatbot extracting correct data based on a given question
- Coding Agent Evaluation
 - Does the coding agent generate correct code and is able to handle constraints
- Human Evaluation
 - Evaluation by experts in various fields both climate experts and non experts

Metrics for RAG evaluation – Faithfulness

- Faithfulness metric : Measure what percentage of the answer generated by a RAG system/chatbot is present in the source documents or contexts
- Claims in the answer and the context can be generated using the LLM.
- Generally a trustworthy metric as the generated claims can be manually evaluated also

$$\text{Faithfulness Score} = \frac{\text{Number of Claims in the generated answer that can be inferred from the context}}{\text{Total number of claims in the generated answer}}$$

Metrics for RAG evaluation – Answer Relevancy

$$\text{answer relevance} = \frac{1}{N} \sum_{i=1}^N \cos(E_{g_i}, E_o)$$

- It helps to understand if the answer generated by the LLM is relevant to the question.
- It can be calculated in two ways
 - Direct cosine similarity between the LLM answer and the question
 - LLM generates possible questions from the answer generated by the chatbot and then cosine similarity is calculated between the generated questions and the original questions.

Coverage Metric

$$\text{Coverage metric} = \left| \frac{\text{common keywords, entities and numbers}}{\text{keywords, entities, numbers in groundtruth}} \right|$$

Keywords matching : Instead of using LLMs to find factual accuracies we can compare the keywords (common to the groundtruth) in both answers and use LLMs to augment the choice of keywords. This metric will help to understand which answer contains the greater number of relevant keywords.

Help to find common keywords and prevent hallucination by the LLMs.

- Helps to avoid problems between Generic and Specific answers
- Possible to manually check the keywords

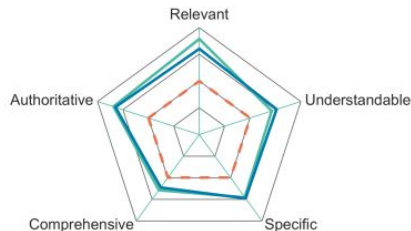
RAG evaluation Results



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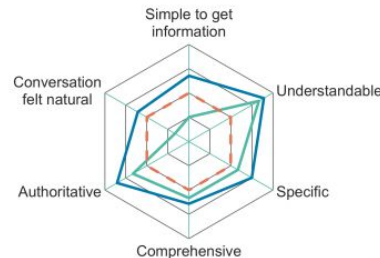
Model Variation	Phase	Faithfulness Mean Score (Sources)	Relevance Mean Score	Coverage Mean Score	Answers with Coverage>ChatGPT	ChatGPT Mean Coverage
F10	1	0.90	0.93	0.34	79.00%	0.26
P10	1	0.89	0.94	0.34	78.00%	0.26
F5	1	0.89	0.93	0.32	72.00%	0.26
P5	1	0.86	0.93	0.32	77.00%	0.26
H20	2	0.81	0.93	0.35	78.00%	0.26
S2BH15	2	0.91	0.94	0.36	85.00%	0.26
S2BH20	2	0.92	0.94	0.36	83.00%	0.26
S2B	2	0.92	0.94	0.34	81.00%	0.26
S2BH-CHR	3	0.93	0.96	0.38	87.00%	0.26
S2BH-DVR	3	0.91	0.94	0.34	84.00%	0.26
S2BH-LST	3	0.90	0.93	0.36	82.00%	0.26
S2BH-REC	3	0.92	0.95	0.35	86.00%	0.26
S2BH-CHR-MQG3	4	0.91	0.90	0.32	81.00%	0.26
S2BH-CHR-MQG5	4	0.92	0.90	0.36	77.00%	0.26

— S2BH-CHR — S2BH-CHR-MQG3 — Neutral Line



(a) Average human ratings of answer quality (n=10 participants). Scaled from strong disagree (inner) to strong agree (outer).

— S2BH-CHR — S2BH-CHR-MQG3 — Neutral Line



(b) Average human ratings of interaction quality (n=10 participants). Scaled from strong disagree (inner) to strong agree (outer).

Figure 3: Human evaluation of answer quality and interaction quality for RAG pipelines SB2H-CHR and SB2H-CHR-MQG3.

Coding Agent Evaluation

- Climate datasets were sampled randomly and synthetic questions corresponding to 3 tasks were generated. This process was repeated 50 times to get 150 questions.
 - Splicing and Retrieval
 - Statistical and Analytical
 - Climate Service and decision support
- Evaluated using LLM as a judge on metrics of
 - Correctness : if the code performs the correct seq of operations
 - Usefulness : How helpful is the code to solve the question
- DA-EVAL : A benchmark dataset of 257 questions and datasets with answers provided.

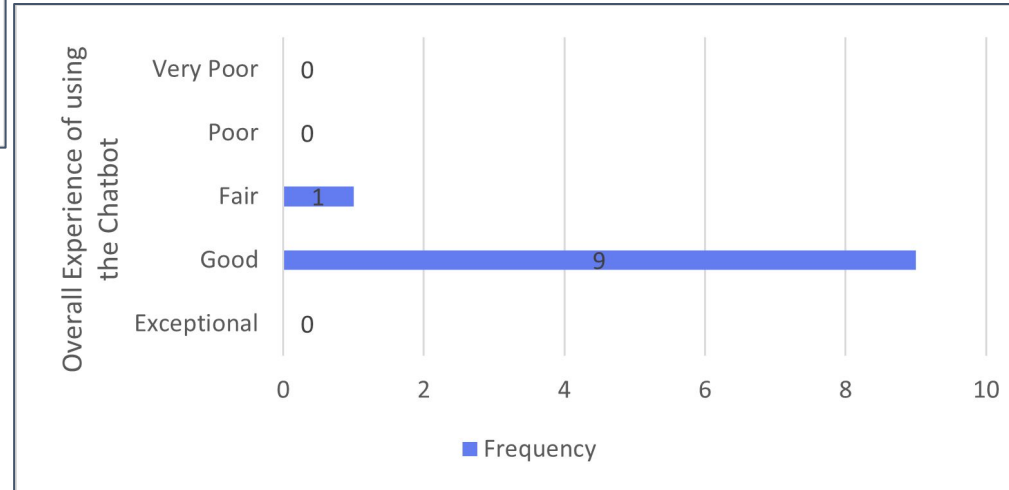
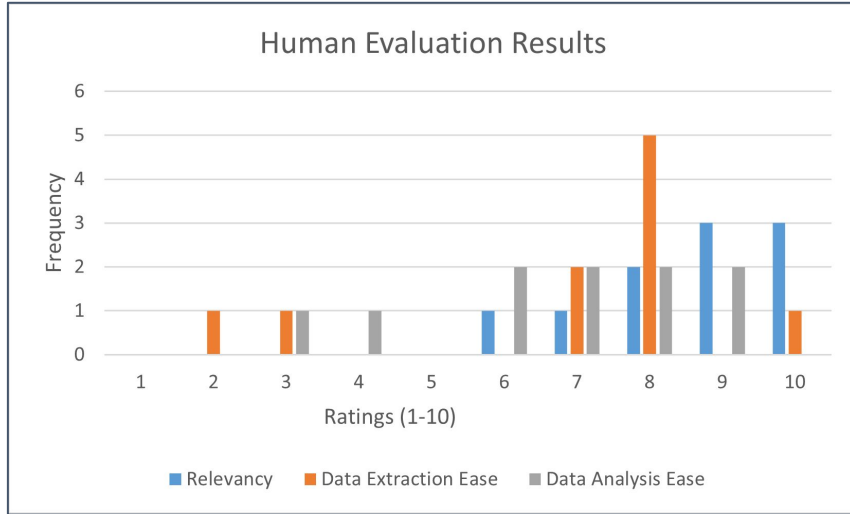
Coding Agent Evaluation



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Dataset	Question Type/Difficulty	Functional Correctness	Usefulness	Correct Answers	Incorrect Answers
Synthetic QA	Splicing and retrieval	3.96	3.96	–	–
	Statistical and analytical	3.96	3.00	–	–
	Climate service and decision support	3.98	4.00	–	–
DA Eval	Easy	3.90	3.80	74	8
	Medium	3.87	3.87	79	8
	Hard	3.95	3.94	76	12

Human Evaluation



Conclusion

- Proposed multi-agentic system is able to communicate information to various end-users
- It can both act as a decision support system for specialists as well as communicate information simply to end users
- Future work would include
 - Increasing the scope of the study to entire europe climate datasets.
 - Connect this multi-agentic system with other systems to create a network of smart agents.

Libraries used

- OpenAI for LLMs (GPT-4o, GPT-4.1, GPT-o1-mini)
- Huggingface for access to embedding models
- LangGraph for coding agent architecture

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